

Ido Lavion

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Education

Israel's Open University

Predicted 2024-2027

B.Sc. in Physics (Currently in 2nd year)

- **GPA:** 93

Hof Hasharon High School

2024-2027

- **GPA:** 10th grade: 97, 11th grade: 93.5
- **Coursework:** 10 units in Multitech (Mechatronics, Mechanics, CAD), 5 units Physics, 5 units Mathematics, 5 units English.
- **Academy in High School Program (2024-2026):** Accelerated Physics and Mathematics tracks to complete requirements early through courses at university.
- **Bioinformatics and Biotechnology Program:** 5 units in computational biology via the Davidson Institute of Science and Technology (Weizmann Institute of Science). GPA: 10th grade: 100, 11th grade: 97.

Work Experience

Private Tutor

2023-2024

- Tutored upto 6 elementary students concurrently in mathematics, Hebrew, English, and sciences.

Martial Arts Instructor

Summer 2024

- Taught self-defense and discipline at a local summer camp for children aged 6-12.
- Organized and developed lesson plans, managing a group of 20-30 children alongside one junior assistant.

Projects

Protein Folding Prediction Using Machine Learning

2026

- Developed a machine learning model to predict protein structures (specifically beta-barrels) from amino acid sequences.
- Completed as the 11th-grade final project for the Bioinformatics and Biotechnology program.

Thrust Vector Control (TVC) Rocket

2026

- Designed a TVC rocket that is theoretically capable of controlled flight for low-altitude operations for the Multitech program.
- Successfully generated 3D models of rocket components, completed the physical design, designed the TVC control system and executed relevant calculations for the compliant TVC mechanism.

Prey and Predator Simulation

2025

- Developed a simulation of prey and predator populations utilizing PPO reinforcement learning.
- Completed as the 10th-grade final project for the Bioinformatics and Biotechnology program.

Skills & Interests

- **Programming:** Python, Java, Assembly (x86), C/C++ (Classical & Microcontrollers), MATLAB, Mathematica, TypeScript, HTML, CSS, SQL.
- **Software:** OnShape, Blender, Unity, Visual Studio/VS Code, Arduino IDE, Adobe Creative Suite, Microsoft Office, LaTeX.
- **Languages:** Hebrew (Native), English (Fluent), Arabic (Basic reading), Japanese (intermediate speaking).
- **Interests:** Physics, Mathematics, Astronomy, Space Exploration, Programming, Robotics, 3D Modeling & Printing, Martial Arts.